

- 2 The reaction between an acid and an alkali is exothermic. You will carry out a neutralisation experiment to determine the enthalpy change involved.

You will mix different volumes of an acid with a fixed volume of an alkali and measure the temperature rises that occur.

FB 4 is aqueous sodium hydroxide, NaOH.

FB 5 is 2.00 mol dm^{-3} hydrochloric acid, HCl.

(a) Method

- Use the thermometer to measure the initial temperature of **FB 4**. Record this initial temperature in the space for results.
- Support the cup in the 250 cm^3 beaker.
- Fill one burette with **FB 5**. Label the burette **FB 5**.
- Fill the other burette with distilled water.

Experiment 1

- Use the 10 cm^3 pipette to transfer 10.0 cm^3 of **FB 4** into the cup.
- Add 9.00 cm^3 of distilled water from the burette to the same cup.
- Add 1.00 cm^3 of **FB 5** from the other burette to the same cup.
- Stir the mixture and use the thermometer to measure the maximum temperature. If necessary, tilt the cup so that the solution covers the bulb of the thermometer.
- Record the maximum temperature in Table 2.1.
- Empty, rinse and dry the cup ready for use in further experiments.

Further experiments

Repeat this method for Experiments 2–5, using 10.0 cm^3 of **FB 4** and the volumes of water and **FB 5** shown in Table 2.1. In each case, measure and record the maximum temperature.

Carry out **two** further experiments, Experiments 6 and 7, which will enable you to determine more precisely the volume of **FB 5** that gives the largest maximum temperature. Record your measurements in Table 2.1.

Results

initial temperature of **FB 4** = °C

Table 2.1

experiment	volume of water / cm^3	volume of FB 5 / cm^3	maximum temperature / °C
1	9.00	1.00	
2	7.00	3.00	
3	5.00	5.00	
4	3.00	7.00	
5	1.00	9.00	
6			
7			

I	
II	
III	
IV	

[4]

